7.7 APPLICATIONS OF AVOGADRO'S LAW

i. It explains Gay-Lussac's law.

ii. It helps in the determination of atomicity of gases.

iii. Molecular formula of gases can be derived using Avogadro's law

iv. It determines the relation between molecular mass and vapour density.

v. It helps to determine gram molar volume of all gases (i.e, 22.4 litre at S.T.P)

7.8 RELATIONSHIP BETWEEN VAPOUR DENSITY AND RELATIVE MOLECULAR MASS

i. Relative molecular mass: (Hydrogen scale)

The Relative Molecular Mass of a gas or vapour is the ratio between the mass of one molecule of the gas or vapour to mass of one atom of Hydrogen.

ii. Vapour Density:

Vapour density is the ratio of the mass of a certain volume of a gas or vapour, to the mass of an equal volume of hydrogen, measured under the same conditions of temperature and pressure.

$$\text{Vapour Density (V.D.)} = \frac{\text{Mass of a given volume of gas or vapour at S.T.P.}}{\text{Mass of the same volume of hydrogen}}$$

According to Avogadro's law, equal volumes of all gases contain equal number of molecules.

Thus, let the number of molecules in one volume = n, then

$$\text{V.D. at S.T.P} = \frac{\text{Mass of 'n' molecules of a gas or vapour at S.T.P.}}{\text{Mass of 'n' molecules of hydrogen}}$$

Cancelling 'n' which is common, you get

$$\text{V.D.} = \frac{\text{Mass of 1 molecule of a gas or vapour at S.T.P.}}{\text{Mass of 1 molecules of hydrogen}}$$

However, since hydrogen is diatomic

$$\text{V.D.} = \frac{\text{Mass of 1 molecule of a gas or vapour at S.T.P.}}{\text{Mass of 2 atoms of hydrogen}}$$

When you compare the formula of vapour density with relative molecular mass, they can be represented as

$$\text{V.D.} = \frac{\text{Mass of 1 molecule of a gas or vapour at S.T.P.}}{2 \times \text{Mass of 1 atom of hydrogen}} \quad \text{(Eqn 7.1)}$$

$$\text{Relative molecular mass (hydrogen scale)} = \frac{\text{Mass of 1 molecule of a gas or vapour at STP}}{\text{Mass of 1 atom of hydrogen}} \quad \text{(Eqn 7.2)}$$

You can therefore substitute the above equation to an Eqn 7.1 and arrive at the following formula

$$\text{V.D.} = \frac{\text{Relative molecular mass}}{2}$$

Now on cross multiplication, you have

2 × vapour density = Relative molecular mass of a gas

(Or)

Relative molecular mass = 2 × Vapour density

7.9 SOLVED PROBLEMS

I. Calculation of molecular mass

Calculate the gram molecular mass of the following.

1) H_2O 2) CO_2 3) $Ca_3(PO_4)_2$

Solution:

1) H_2O

Atomic masses of H = 1, O = 16

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